Observational Response Surfaces versus Interventional Policy: A Critical Audit of Causal Discovery and Multi-Objective Optimization Across Indian Urban Air Quality Networks

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Abstract

The uncritical application of machine learning and multi-objective optimization to ambient air-quality telemetry can produce spurious, policy-shaped conclusions, because non-linear atmospheric chemistry and strong spatio-temporal autocorrelation violate the assumptions on which naive pipelines rest. We audit a complete prediction-to-policy pipeline built on 204 continuous ambient monitoring stations across urban India—1,180,551 station-hours of criteria pollutants ($PM_{2.5}$, PM_{10}, NO, NO_2, SO_2, CO, O_3) over calendar year 2025, merged with collocated Open-Meteo reanalysis meteorology—and report three findings. First, spatio-temporal autocorrelation inflates apparent skill under random row splitting: contemporaneous ozone estimation attains $R^2 \approx 0.41$ under an i.i.d. random split but only $R^2 \approx 0.25$ under a strict blocked temporal (Q4) holdout and $R^2 \approx 0.30$ under spatial leave-stations-out validation, while a previous-hour persistence baseline reaches $R^2 \approx 0.79$; pollutants contribute only $\Delta R^2 \approx +0.12$ beyond meteorology. Second, linear structural discovery (DirectLiNGAM) recovers a single bootstrap-stable, meteorology-robust edge consistent with *localized* nitric-oxide titration ($NO + O_3 \rightarrow NO_2 + O_2$) rather than any regional photochemical-smog mechanism, a limitation sharpened by the structural absence of volatile organic compounds (VOCs) and by an authoritative unit audit showing CO, NO_2 and SO_2 are reported in mixed ppb/$\mu g\,m^{-3}$ units with no per-record provenance. Third, we prove analytically that over the linear response surface the true Pareto frontier collapses to a one-dimensional segment, against which 100% of the heuristic NSGA-II candidate policies are dominated (generational distance = 0.084; hypervolume error = 6.6%). We reframe the artifact as an *open, leakage-safe benchmark* and propose a four-part scientific identification gate for observational environmental AI.

Keywords: air quality; data leakage; causal discovery; DirectLiNGAM; multi-objective optimization; Pareto dominance; source apportionment; reproducible benchmark; India.

1 Introduction

Urban air pollution is among the most severe environmental health burdens in South Asia. Rapid urbanization and motorization across the Indo-Gangetic Plain and the peninsular metropolitan corridors have driven chronic and widespread exceedances of the National Ambient Air Quality

Standards (NAAQS), with annual-mean fine particulate matter ($PM_{2.5}$) across many Indian cities exceeding national limits by large multiples and the World Health Organization guideline by an order of magnitude [8, 20]. The expansion of the Continuous Ambient Air Quality Monitoring Station (CAAQMS) network operated under the Central Pollution Control Board, together with open aggregation platforms such as OpenAQ [18], has made hourly, multi-pollutant telemetry available at a national scale. This abundance of observational data has, in turn, invited an increasingly ambitious class of data-driven analyses that chain together three steps: *prediction* of ambient concentrations, *discovery* of a causal structure among pollutants, and *optimization* of "policies" framed as pollutant reductions.

1.1 Mechanistic models versus observational learning

Two methodological traditions sit in tension. Chemical transport models (CTMs) encode the governing physics and chemistry of the atmosphere and can, in principle, answer genuinely interventional questions—what happens to concentrations if a given emission sector is curtailed. In practice their skill is bounded by bottom-up emission inventories that are frequently incomplete, out of date, or spatially mis-resolved, particularly over rapidly changing Indian source regions [8]. Observational machine learning inverts this trade-off: methods such as gradient-boosted trees [3] and random forests [2] excel at fitting the empirical covariation of pollutants and meteorology, but they estimate *associational response surfaces*, not mechanisms. The central methodological risk is that the fluency of the fit is mistaken for mechanistic or interventional content.

1.2 The ozone–NO_x paradox

Nowhere is this risk sharper than for ozone. Tropospheric ozone is a secondary pollutant produced by the photochemical oxidation of volatile organic compounds (VOCs) and carbon monoxide in the presence of nitrogen oxides ($NO_x = NO + NO_2$) and sunlight [17, 26]. Its production rate is a famously non-linear, non-monotone function of precursor concentrations. In the *NO_x-limited* regime, typical of rural and background environments, ozone increases with NO_x; in the *VOC-limited* (or NO_x-saturated) regime, typical of fresh urban emissions, additional NO_x *suppresses* ozone through hydroxyl-radical (OH) scavenging and direct titration, $NO + O_3 \rightarrow NO_2 + O_2$ [29]. Which regime holds at a given site and hour depends on the local VOC-to-NO_x ratio. A single pooled, linear, contemporaneous model fitted across hundreds of heterogeneous urban stations cannot represent this regime dependence; at best it recovers a spatial and temporal average dominated by the urban titration signature. Any apparent "mechanism" it reports is therefore constrained—and potentially confounded—before a single parameter is estimated, especially when VOCs are not measured at all.

1.3 The concentration–emission conflation

A further, and more consequential, conflation underlies the "optimization" step. A response surface relates *ambient concentrations* to one another. An actual air quality policy acts on *sectoral emissions*, which map to concentrations only through transport, chemistry and the diurnally varying planetary boundary layer. An optimizer that searches over standardized ambient concentrations and returns a vector such as "set NO_2 to -2σ" is not recommending an emission intervention: it is selecting a point on a correlational surface that may be physically unreachable, and whose objective values carry no policy-benefit interpretation. Treating such a point as a control recommendation is a category error. The distinction between predicting concentrations and simulating interventions is the modern, data-driven instance of the long-standing gap between observation and intervention formalized in the causal-inference literature [21, 22].

1.4 Related work

Recent air-quality machine learning has advanced rapidly on prediction while, we argue, under-addressing evaluation and causal rigour. On the predictive side, deep-learning architectures now dominate: convolutional autoencoders reconstruct $PM_{2.5}$ fields over India [24], and hybrid CNN–RNN models report strong short-horizon skill [31]; recent systematic reviews confirm that this literature "prioritizes accuracy over interpretability" [16]. Crucially, most such studies still evaluate with random splits of autocorrelated data; Just et al. [11] demonstrated, for satellite-based $PM_{2.5}$, that spatial cross-validation withholding nearby sites yields substantially higher (honest) error than conventional random splitting—the precise leakage we quantify here. On the causal and explainable side, convergent cross-mapping [23] and causal-forest/XAI hybrids [33] have begun to probe $PM_{2.5}$–ozone and ozone-rebound drivers, consistently finding meteorology dominant; and SHAP-based "de-weathering" of ozone trends [34] explicitly flags that attribution is unstable under multicollinearity. On source apportionment, receptor modelling continues to require chemical speciation and uncertainties that criteria-pollutant telemetry lacks [13, 19]. Our work is complementary and corrective: we apply these same method families but foreground the diagnostics they usually omit—leakage-safe protocols, bootstrap edge stability, explicit meteorological de-confounding, an exact-versus-heuristic optimality check, and an authoritative unit-provenance audit—and package the result as an open benchmark.

1.5 Contributions

Rather than add another over-claimed prediction-to-policy pipeline, we audit one. We built the complete three-stage system—clustering, prediction, causal discovery, and multi-objective optimization with post-hoc explanation—ran it on real national-scale telemetry, and then subjected every claim to adversarial, reproducible scrutiny. Our contributions are:

1. **An empirical data and quality-control audit** of 204 CAAQMS stations (1,180,551 station-hours, 2025), including an authoritative per-sensor unit verification that reveals a genuine ppb/μg m^{-3} unit mix for CO, NO_2 and SO_2 with no per-record provenance (Section 2).

2. **A leakage-safe predictive evaluation** that contrasts i.i.d. random splitting with blocked temporal and spatial leave-stations-out protocols and a full baseline ladder, quantifying how autocorrelation inflates apparent skill (Sections 3 and 4).

3. **A structural-equation analysis** via DirectLiNGAM with bootstrap edge stability and explicit meteorology-confounding tests, isolating the single robust, chemically interpretable association (Section 4).

4. **An analytical proof of the global Pareto frontier** for the fitted linear surrogate (Lemma 1), against which we benchmark the heuristic NSGA-II solutions and quantify their complete dominance.

5. **A scientific identification gate and an open, reproducible benchmark** for observational environmental AI (Sections 5 and 6).

2 Data and Quality Control

2.1 Network and acquisition

We assembled hourly observations for the seven criteria pollutants—$PM_{2.5}$, PM_{10}, NO, NO_2, SO_2, CO and O_3—from 204 continuous ambient monitoring stations across urban India for calendar year 2025, retrieved through the OpenAQ v3 API [18], which aggregates and standardizes

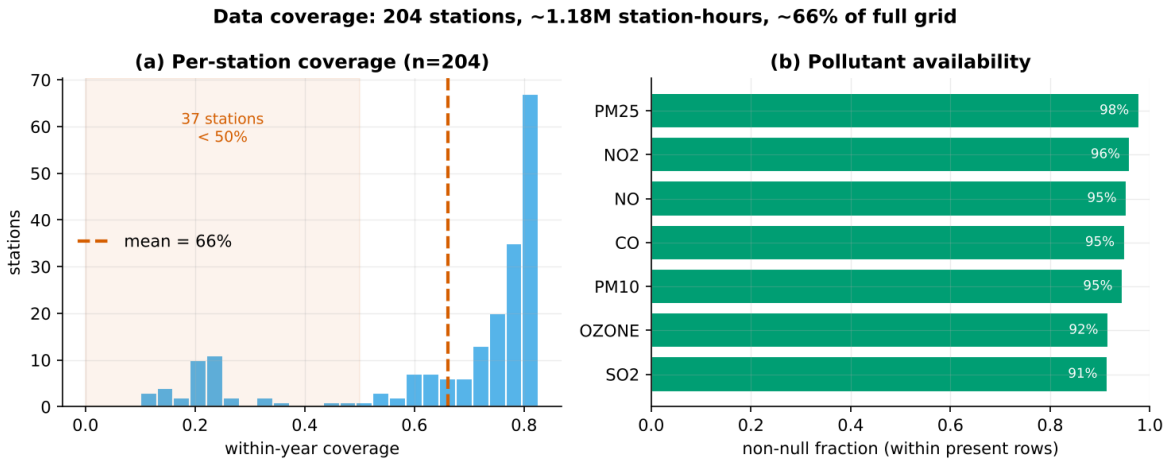


Figure 1: Network coverage. (a) Distribution of within-year temporal completeness across the 204 stations (mean 66%; 37 stations below 50%, shaded). (b) Per-pollutant non-null fraction within present station-hours. Geolocation of individual monitors is not retained in the public telemetry export and is therefore not mapped; the network is urban and spatially concentrated in the National Capital Region and major metropolitan centres (a representativeness limitation, Section 5).

the regulatory CAAQMS/CPCB feeds. Each station was merged, at hourly resolution, with collocated Open-Meteo reanalysis meteorology [36]: 2 m air temperature, relative humidity, 10 m wind speed, and surface pressure, sampled at the nearest ∼0.1° reanalysis grid cell and cached per coordinate to avoid redundant retrieval. All timestamps are stored in UTC; because the network spans a single nominal time zone (Indian Standard Time, UTC+5:30), diurnal features are computed after conversion to local solar-aligned hours. The assembled corpus comprises 1,180,551 station-hours.

2.2 Coverage and completeness

The realized corpus covers approximately 66% of a complete 204 × 8,760 station-hour grid (Figure 1a). Completeness is markedly heterogeneous across sites: the median within-year coverage is 0.77, but a left tail of 37 stations falls below 50%. Missingness is a mixture of structural (no sensor deployed for a given species), intermittent (sensor dropout), and below-detection values that are not separable in the absence of detection-limit metadata. Per-species availability within present rows (Figure 1b) is high for $PM_{2.5}$ (0.98) and the gases (0.91–0.96). Because autocorrelated gaps interact dangerously with random splitting (Section 3), all predictive evaluation uses blocked or grouped protocols rather than imputing across gaps.

2.3 Sanitization

Three classes of non-physical artifact were identified and flagged. (i) *Instrument saturation / round caps*: distinct spikes of mass at round ceilings (NO, NO_2, O_3 near 500; $PM_{2.5}$ near 1000). (ii) *Zero-padding and dropout*: runs of exact zeros and abrupt gaps inconsistent with ambient variability. (iii) *Non-physical outliers*: 35 values exceeding physically plausible bounds were flagged and excluded from distributional summaries. Screening these artifacts materially changes downstream covariance estimates—for example, the CO–NO correlation rises from 0.14 to 0.42—which is itself evidence that naive clustering is sensitive to data hygiene (Section 4).

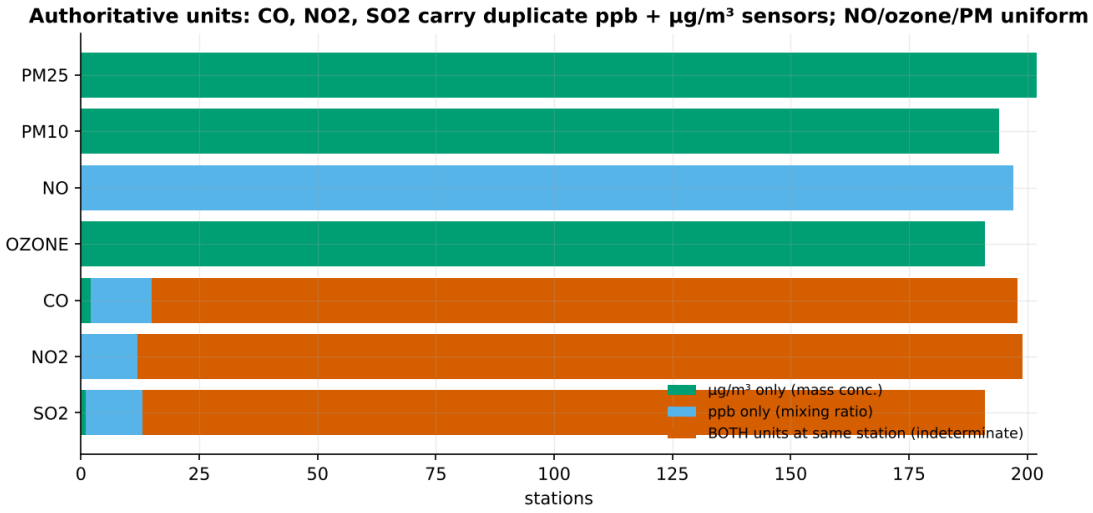


Figure 2: Authoritative unit verification per species. $PM_{2.5}$, PM_{10} and O_3 are uniform $\mu g\,m^{-3}$; NO is uniform ppb; CO, NO_2 and SO_2 carry duplicate ppb and $\mu g\,m^{-3}$ sensors at the same station (orange), rendering their stored unit indeterminate.

2.4 The unit-provenance discovery

A pre-registration-style verification of measurement units against the *authoritative* OpenAQ per-sensor metadata revealed a material provenance defect (Figure 2). NO is reported in parts-per-billion (ppb) at every station; O_3, $PM_{2.5}$ and PM_{10} are reported in mass concentration ($\mu g\,m^{-3}$) and are unit-consistent. However, CO, NO_2 and SO_2 carry *duplicate sensors reporting in both ppb and* $\mu g\,m^{-3}$ at almost every station (CO at 183/198 stations, NO_2 at 187/199, SO_2 at 178/191), while the exported telemetry retains a single value per pollutant per station-hour with *no record of which sensor produced it.* A mixing ratio (ppb) and a mass concentration ($\mu g\,m^{-3}$) are related only through molecular weight and ambient temperature and pressure; they are not comparable, and the mix cannot be reconciled post hoc from the stored data. This supersedes an earlier, value-based inference (which had suggested a uniform CO regime): the authoritative sensor metadata governs. Consequently, we restrict all quantitative cross-station inference to the unit-consistent species (NO, O_3, $PM_{2.5}$, PM_{10}); the two headline results below—ozone estimation and the NO→O_3 association—depend only on these.

2.5 Summary statistics

Table 1 reports distributional summaries of the sanitized telemetry. Values are in the stored measurement scale; for the unit-indeterminate species (CO, NO_2, SO_2) the magnitudes are not cross-station comparable and are shown for completeness only. The heavy right skew of $PM_{2.5}$ and PM_{10} (large standard deviations, high 95th percentiles) is characteristic of the Indian urban winter particulate regime.

3 Methodology

3.1 Governing physical dynamics

The ambient concentration field $C_i(\mathbf{x}, t)$ of chemical species i evolves according to the atmospheric advection–diffusion–reaction equation,

$$\frac{\partial C_i}{\partial t} = \underbrace{-\nabla\cdot(\mathbf{u}\,C_i)}_{\text{advection}} + \underbrace{\nabla\cdot(\mathbf{K}\,\nabla C_i)}_{\text{turbulent diffusion}} + \underbrace{R_i(C_1,\ldots,C_m,\,T,\,h\nu)}_{\text{chemistry}} + \underbrace{E_i(\mathbf{x},t)}_{\text{emissions}}, \tag{1}$$

Table 1: Summary statistics of the sanitized telemetry (calendar year 2025, 204 stations). Missing fraction is relative to present station-hours. CO, NO_2, SO_2 are unit-indeterminate (Section 2.4); PM in $\mu g\,m^{-3}$, O_3 in $\mu g\,m^{-3}$, NO in ppb.

Pollutant	Mean	Std. dev.	5th pct.	95th pct.	Missing frac.
O_3	30.52	30.89	2.57	89.80	0.084
NO	14.81	29.13	0.88	49.20	0.047
NO_2	27.33	27.37	2.60	80.10	0.040
CO	0.87	2.40	0.10	2.18	0.051
SO_2	13.47	25.67	1.75	35.20	0.085
PM_{10}	121.50	116.48	18.20	349.20	0.055
$PM_{2.5}$	58.28	73.79	6.50	185.00	0.022

where $\mathbf{u}(\mathbf{x}, t)$ is the wind velocity field, $\mathbf{K}$ the (second-order) eddy-diffusivity tensor, R_i the net chemical production/loss rate for species i as a function of all m reacting species, temperature T and the actinic photon flux $h\nu$, and E_i the surface emission rate. Equation (1) makes the epistemic gap explicit: a single-site, contemporaneous regression of C_i on the other C_j and a few meteorological scalars discards the spatial transport operators entirely, linearizes the strongly non-linear reaction term R_i, and—critically— never observes the emission field E_i on which policy acts. It can estimate a local, time-averaged *response surface*, not the operator in (1).

3.2 Predictive formulation and leakage controls

Let $y_{s,t}$ denote the standardized ozone concentration at station s and hour t, and $\mathbf{x}_{s,t} \in \mathbb{R}^d$ the vector of contemporaneous covariates (the other six pollutants and four meteorological variables). We estimate $\hat{y} = f(\mathbf{x})$ with gradient-boosted regression trees [3], chosen for their strong tabular performance under non-linear interactions and implemented within a distributed Apache Spark pipeline [35]; hyper-parameters are fixed a priori and never tuned on any evaluation fold.

Because $y_{s,t}$ is strongly autocorrelated in both time and space, the choice of evaluation protocol is not a detail but the central inferential control [12, 25, 32]. We contrast three protocols:

1. *i.i.d. random split* (70/30 over rows): the standard but *leaky* baseline, in which near-duplicate neighbouring hours of the same station appear in both train and test.
2. *Blocked temporal holdout*: train on quarters Q1–Q3 (months 1–9), evaluate on Q4 (months 10–12). Temporal blocks are never split.
3. *Spatial leave-stations-out*: 5-fold `GroupKFold` with the station identifier as the grouping key, so that no station appears in both train and test; this probes transfer to *unmonitored* locations.

All preprocessing (standardization, imputation) is fit on the training fold only. We report against a ladder of reference baselines: (i) *lag-1 persistence*, $\hat{y}_{s,t} = y_{s,t-1}$; (ii) a *station* $\times$ *hour* $\times$ *month diurnal-seasonal mean*, the strongest purely climatological predictor; and (iii) an L_2-regularized *Ridge* regression [9] on the same covariates, as a linear reference for the boosted model. Uncertainty is quantified by a station-block bootstrap.

3.3 Covariance clustering versus bilinear receptor modelling

The pipeline's first stage groups species by k-means over their inter-species Pearson correlation profiles. It is essential to state precisely what this does *not* do. Quantitative source apportionment

is a bilinear receptor model; positive matrix factorization (PMF) [19] factorizes a speciated concentration matrix $X = [x_{ij}]$ (sample i, chemical species j) as

$$x_{ij} = \sum_{k=1}^{p} g_{ik} f_{kj} + e_{ij}, \qquad g_{ik} \geq 0,\ f_{kj} \geq 0, \tag{2}$$

where g_{ik} is the contribution of source k to sample i, f_{kj} the chemical profile of source k, and e_{ij} a residual weighted by the measurement uncertainty σ_{ij} through the objective $Q = \sum_{ij}(e_{ij}/\sigma_{ij})^2$. PMF requires (a) chemically speciated composition (including organic and elemental carbon and key tracers), (b) per-measurement uncertainties, and (c) enough species to identify p non-negative factors. Our data provide only seven criteria pollutants, no speciation, no VOCs, and no per-measurement uncertainties. k-means over criteria-gas correlations is therefore an *exploratory inter-species covariance clustering*, and its clusters are mathematically distinct from the latent source factors $\{g_{ik}, f_{kj}\}$ of (2); they admit no mass-balance or source interpretation.

3.4 Causal discovery via DirectLiNGAM

For exploratory structure, we model the standardized pollutant vector $\mathbf{z} \in \mathbb{R}^m$ as a linear, non-Gaussian, acyclic structural equation model,

$$\mathbf{z} = \mathbf{B}\,\mathbf{z} + \mathbf{e}, \tag{3}$$

where $\mathbf{B}$ is a strictly lower-triangular (under some causal ordering) matrix of structural coefficients and $\mathbf{e}$ is a vector of mutually independent, non-Gaussian exogenous disturbances. Unlike constraint-based discovery [30], DirectLiNGAM [27, 28] estimates a causal ordering and the coefficients $\mathbf{B}$ directly, exploiting non-Gaussianity to orient edges; non-linear generalizations exist via additive-noise models [10] but inherit the same confounding sensitivity. Its validity rests on four assumptions: (A1) *acyclicity* of the generating graph; (A2) *linearity* of the structural equations; (A3) *non-Gaussian* disturbances with non-zero variance; and (A4) *causal sufficiency*—no unobserved common causes of two or more observed variables. We encode domain knowledge as a prior-knowledge constraint matrix $K \in \{0, 1, -1\}^{m \times m}$ that forbids edges from secondary pollutants (e.g. O_3) into primary combustion tracers (e.g. NO, CO), consistent with emission chemistry. Edge reliability is assessed by a station-block bootstrap (reporting per-edge selection frequency and coefficient sign), and—decisively—by refitting with the meteorological variables included as exogenous nodes, which tests assumption (A4) against the dominant confounder.

3.5 Analytical derivation of the global Pareto frontier

The optimization stage searches, with the NSGA-II evolutionary algorithm [5] as implemented in the Optuna framework [1], for policies minimizing two standardized objectives built from the fitted linear response surface. Empirically the fitted surface is

$$z_{O_3} = \alpha\, z_{NO}, \qquad \alpha < 0, \tag{4}$$

$$z_{PM_{2.5}} = \beta_1 z_{NO} + \beta_2 z_{NO_2} + \beta_3 z_{CO}, \qquad \beta_1, \beta_2, \beta_3 > 0, \tag{5}$$

with fitted values $\alpha = -0.124$, $\beta_1 = 0.092$, $\beta_2 = 0.219$, $\beta_3 = 0.228$; SO_2 and VOC enter with zero coefficient (VOC is not a fitted variable), so the effective decision vector is $\mathbf{z} = (z_{NO}, z_{NO_2}, z_{CO})$ over the search hypercube $z_j \in [z_{\min}, z_{\max}] = [-2, 0]$ (reductions only, bounded at -2σ).

Lemma 1 (Collapse of the linear Pareto front)**.** *Consider the bi-objective minimization of* $\big(f_1(\mathbf{z}), f_2(\mathbf{z})\big) = \big(z_{O_3}, z_{PM_{2.5}}\big)$ *defined by* (4)–(5) *over* $\mathbf{z} \in [z_{\min}, z_{\max}]^3$ *with* $\alpha < 0$ *and* $\beta_1, \beta_2, \beta_3 > 0$*. Then every Pareto-optimal point satisfies* $z^\star_{NO_2} = z^\star_{CO} = z_{\min}$*, and the Pareto set is the one-dimensional segment*

$$\mathcal{P} = \Big\{ \big(z_{NO}, z_{\min}, z_{\min}\big) \ :\ z_{NO} \in [z_{\min}, z_{\max}] \Big\},$$

whose image is the line segment $z_{\mathrm{PM}_{2.5}} = \frac{\beta_1}{\alpha}\, z_{\mathrm{O}_3} + (\beta_2 + \beta_3)\, z_{\min}$.

Proof. The objective f_1 depends only on z_{NO}, while f_2 is strictly increasing in each of z_{NO_2} and z_{CO} (as $\beta_2, \beta_3 > 0$) and does not appear in f_1. Fix any feasible point $\mathbf{z}$ with $z_{\mathrm{NO}_2} > z_{\min}$. The point $\mathbf{z}'$ obtained by lowering z_{NO_2} to $z_{\min}$ (all else equal) is feasible, leaves f_1 unchanged, and strictly decreases f_2; hence $\mathbf{z}'$ dominates $\mathbf{z}$, so $\mathbf{z}$ is not Pareto-optimal. The identical argument applies to z_{CO}. Therefore every Pareto-optimal point has $z^\star_{\mathrm{NO}_2} = z^\star_{\mathrm{CO}} = z_{\min}$.

Restricted to this face the objectives reduce to $f_1 = \alpha z_{\mathrm{NO}}$ and $f_2 = \beta_1 z_{\mathrm{NO}} + (\beta_2 + \beta_3) z_{\min}$. Since $\alpha < 0$, f_1 is strictly decreasing in z_{NO}, whereas f_2 is strictly increasing in z_{NO} (as $\beta_1 > 0$); the two objectives are therefore in strict conflict along z_{NO}, so no interior point of the segment dominates another and the entire segment is Pareto-optimal. Eliminating $z_{\mathrm{NO}} = f_1/\alpha$ gives $f_2 = \frac{\beta_1}{\alpha} f_1 + (\beta_2 + \beta_3) z_{\min}$, a straight line. □

Corollary 1. *The true Pareto frontier of the fitted linear surrogate is a closed-form line segment computable without any metaheuristic search. Any evolutionary approximation can at best match it and, lacking the boundary structure of Lemma 1, will generically return interior, dominated points.*

Benchmark metrics. Let $\mathcal{F}$ be a dense discretization of the exact frontier $\mathcal{P}$ and $A = \{\mathbf{a}_k\}_{k=1}^n$ the archive of heuristic solutions in objective space. The *generational distance* is

$$\mathrm{GD}(A) = \Big(\frac{1}{n}\sum_{k=1}^{n} d_k^2\Big)^{1/2}, \qquad d_k = \min_{\mathbf{p}\in\mathcal{F}} \|\mathbf{a}_k - \mathbf{p}\|_2, \tag{6}$$

and, for a shared reference point $\mathbf{r}$ dominated by all points, the *hypervolume error* is

$$\mathrm{HVE} = \frac{\mathrm{HV}(\mathcal{F}; \mathbf{r}) - \mathrm{HV}(A; \mathbf{r})}{\mathrm{HV}(\mathcal{F}; \mathbf{r})}, \tag{7}$$

where $\mathrm{HV}(\cdot\,; \mathbf{r})$ is the dominated hypervolume [37]. GD = 0 and HVE = 0 iff the archive lies on the true frontier.

4 Results

4.1 Predictive benchmarking and the cost of leakage

Table 2 reports contemporaneous ozone estimation across the three evaluation protocols and the baseline ladder. The result is unambiguous: apparent skill is an artifact of the protocol. The gradient-boosted model attains $R^2 = 0.412$ under the i.i.d. random split, but this collapses to $R^2 = 0.251$ under the blocked temporal Q4 holdout and $R^2 = 0.304$ under spatial leave-stations-out, falling further to $R^2 = 0.219$ under the strict temporal×spatial intersection (Figure 3). The ∼0.16–0.19 absolute R^2 gap between the random split and the blocked/grouped protocols is pure spatio-temporal leakage: neighbouring hours of the same station, near-duplicates of one another, leak across the random train/test boundary [12, 25].

Two baselines frame the honest skill. The linear Ridge reference is weak everywhere ($R^2 \le 0.17$), confirming that the signal the boosted model captures is genuinely non-linear; but a trivial *lag-1 persistence* predictor attains $R^2 = 0.794$ on the same temporal holdout. The learned model is thus valuable only where persistence is unavailable—namely spatial interpolation to unmonitored sites—and is *beaten by doing nothing* for temporal gap-filling. The diurnal-seasonal climatology ($R^2 = 0.041$) shows that static station behaviour alone explains almost none of the hour-to-hour variance.

Table 2: Leakage-safe predictive benchmarking of contemporaneous ozone estimation. All preprocessing fit on the training fold only; hyper-parameters fixed a priori. Persistence and the diurnal-seasonal mean are defined only for the temporal holdout. Lower RMSE/MAE and higher R^2 are better.

Protocol	Model	R^2	RMSE	MAE
Random split (leaky)	XGBoost	0.412	23.72	14.90
	Ridge (L_2)	0.167	28.21	18.63
Blocked temporal (Q1–Q3 $\rightarrow$ Q4)	XGBoost	0.251	26.49	16.19
	Ridge (L_2)	0.082	29.32	18.78
	Diurnal-seasonal mean	0.041	29.98	20.04
	Persistence (lag-1)	**0.794**	**13.88**	**5.57**
Spatial leave-stations-out (5-fold)	XGBoost	0.304	27.94	18.45
	Ridge (L_2)	0.159	28.33	18.73
Temporal $\times$ spatial (strict)	XGBoost	0.219	30.00	18.41

Table 3: Robustness of ozone estimation to dropping the unit-indeterminate species (CO, NO_2, SO_2). Held-out R^2; XGBoost; preprocessing fit on train only.

Feature set	Blocked temporal	Leave-stations-out
Weather only	0.140	0.214
Weather + unit-consistent (NO, $PM_{2.5}$, PM_{10})	0.210	0.266
Weather + all pollutants	0.253	0.271

4.2 Meteorology ablation

Stripping covariate groups under the blocked temporal protocol isolates their contributions. A weather-only model attains $R^2 = 0.140$ and a pollutant-only model $R^2 = 0.132$, while the full model reaches $R^2 = 0.251$. The incremental value of the six pollutants *beyond* meteorology is therefore only $\Delta R^2 \approx +0.117$: most of the explainable ozone variance is meteorological, and the chemical covariates add a small, though non-trivial, increment. Any downstream "mechanistic" reading must account for the fact that meteorology, not chemistry, carries the bulk of the signal.

4.3 Robustness to the unit-indeterminate species

Because CO, NO_2 and SO_2 are unit-indeterminate (Section 2.4), we re-ran the leakage-safe estimation using *only* the unit-consistent pollutants (NO, $PM_{2.5}$, PM_{10}, plus meteorology), dropping the three mixed species entirely (Table 3). The conclusions are unchanged. Under the blocked temporal protocol, weather plus the unit-consistent pollutants attains $R^2 = 0.210$ versus 0.253 for the full model—a difference of only 0.043—and under leave-stations-out the two are within 0.005 (0.266 vs. 0.271). The incremental pollutant signal beyond meteorology remains small either way (+0.07 unit-consistent, +0.11 all species). The headline—modest honest skill, dominated by meteorology—therefore does not depend on the species whose units cannot be verified. The single stable causal edge, NO $\rightarrow$ O_3 (Section 4), likewise involves only unit-consistent variables (NO in ppb, O_3 in $\mu g\,m^{-3}$) and is unaffected by construction.

4.4 Recovered structure and its atmospheric interpretation

The bootstrap-stabilized DirectLiNGAM structure is shown in Figure 4. Of all candidate edges, exactly one is both bootstrap-stable (selection frequency 1.0) and survives the inclusion of

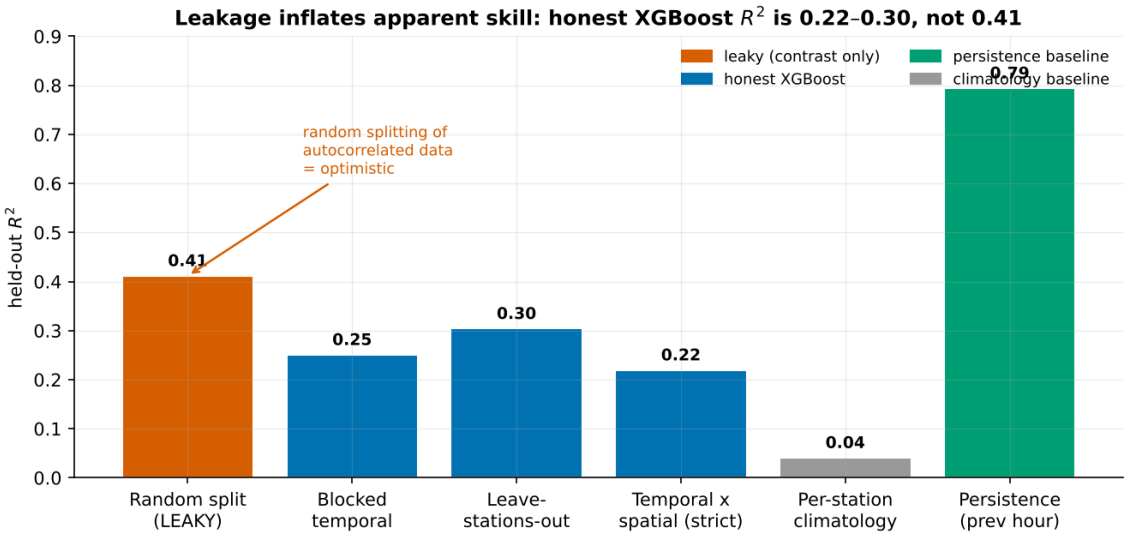


Figure 3: Held-out R^2 for contemporaneous ozone by evaluation protocol. Random splitting of autocorrelated telemetry inflates skill to 0.41; honest blocked and grouped protocols give 0.22–0.30. A previous-hour persistence baseline (green) dominates for temporal gap-filling.

meteorology as exogenous nodes: the negative edge NO $\rightarrow$ O_3, with standardized coefficient -0.138 ± 0.007 in the pollutant-only model, attenuating to -0.108 once temperature, humidity, wind and pressure are included. This is the chemical signature of *localized* titration, $NO + O_3 \rightarrow NO_2 + O_2$, expected in fresh, NO-rich urban (traffic-dominated) air [17, 29]: ambient NO destroys ozone locally. Crucially, this is *not* regional photochemical production, which is governed by VOC oxidation that these data cannot observe. The remaining high-frequency edges (NO $\rightarrow$ NO_2, +0.40; $NO_2 \rightarrow PM_{2.5}$, +0.23; and the CO combustion edges, frequency 0.91) are consistent with shared combustion co-emission but involve unit-indeterminate species (Section 2.4) and, for the particulate edges, collapse once meteorology is included—evidence that the pollutant-only graph is confounded by the boundary layer (Section 5). SO_2 edges are unstable (frequency ≤ 0.5) and are not retained.

4.5 Empirical Pareto frontier versus the NSGA-II archive

Lemma 1 predicts that the true frontier is a one-dimensional segment with $z_{NO_2} = z_{CO} = z_{\min}$. We computed this exact segment in closed form and audited the 19 saved NSGA-II candidate policies against it (Table 4, Figure 5). Every one of the 19 heuristic policies is dominated by the exact frontier—a 100% dominance rate—with a mean objective-space gap of 0.046 (maximum 0.298) in standardized $PM_{2.5}$, a generational distance of GD = 0.084 (6), and a hypervolume error of HVE = 6.6% (7). The heuristic fails precisely as Corollary 1 anticipates: unaware of the boundary structure, NSGA-II scatters interior points rather than driving the reducible coordinates to their bound. For a linear surrogate, the metaheuristic is not merely unnecessary—it is strictly inferior to a closed-form solution that any analyst can write down. We stress that this linearity is not a restriction we imposed: it is the response surface the audited pipeline itself fitted and optimised over, so the dominance is a property of the deployed system, not of a straw-man model. The general lesson is that any optimiser should be checked against an exact or analytic baseline whenever the fitted surrogate admits one.

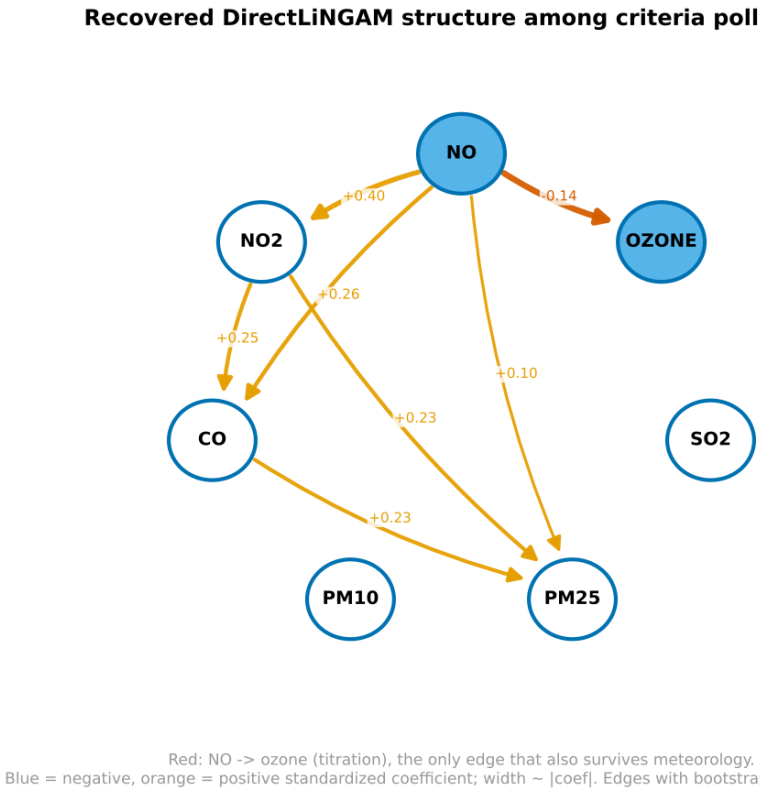


Figure 4: Recovered DirectLiNGAM structure (bootstrap frequency ≥ 0.9). Only NO $\rightarrow$ O_3 (red, negative) is both stable and robust to meteorological confounding. Edge width scales with |coefficient|; SO_2 carries no stable edge.

4.6 Model explainability

Figure 6 shows the global SHAP importance ranking [14, 15] for the ozone model. Relative humidity and temperature dominate, followed by NO as the leading chemical covariate—mirroring both the meteorology ablation and the single stable structural edge. We stress the epistemic status of this agreement: SHAP and DirectLiNGAM are two summaries of the *same* fitted model and data. Their concordance on the sign and prominence of NO is *internal consistency*, not independent validation; it cannot license a causal claim that the leakage-safe and confounding analyses do not already support.

5 Discussion: A Scientific Identification Gate

The audit in Section 4 is not a catalogue of incidental shortcomings; the three findings share a common root. A pooled, linear, observational model of ambient telemetry cannot, in principle, answer the interventional and apportionment questions the original pipeline posed. We formalize this as a four-part *identification gate*: criteria that an observational air-quality analysis must pass before causal, apportionment or policy language is warranted.

5.1 Gate 1: structural absence of VOCs

Ozone sensitivity is governed by the VOC-to-NO_x ratio [17, 26, 29]. Without continuous hydrocarbon speciation—or an observational proxy such as the formaldehyde-to-NO_2 column ratio ($HCHO/NO_2$) from satellite retrievals—the VOC-limited and NO_x-limited regimes are empirically indistinguishable in these data. A model that never observes VOCs cannot resolve

Table 4: Audit of the heuristic NSGA-II archive against the exact analytical Pareto frontier of the fitted linear surrogate (Lemma 1). Objectives are standardized; gaps are Euclidean in objective space.

Quantity	Value
Heuristic candidate policies audited	19
Policies dominated by the exact frontier	19 (100%)
Mean gap to frontier (std. $PM_{2.5}$)	0.046
Maximum gap to frontier	0.298
Generational distance, GD	0.084
Hypervolume error, HVE	6.6%
Exact frontier: $z^\star_{NO_2} = z^\star_{CO}$	$z_{\min} = -2$
Exact frontier endpoints $(z_{O_3}, z_{PM_{2.5}})$	$(0, -0.89) \to (0.25, -1.08)$

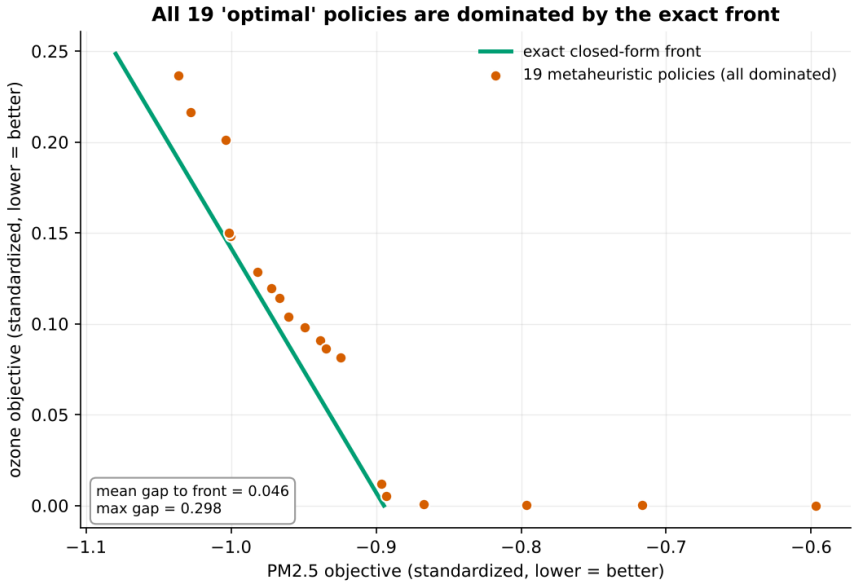


Figure 5: The 19 NSGA-II "optimal" policies (orange) against the exact closed-form Pareto frontier (green) of the linear surrogate. All heuristic points are dominated.

which regime produced a given ozone value, and so cannot attribute an ozone change to a precursor intervention. Our single recovered edge, NO $\to$ O_3 < 0, is consistent with this limitation: it is the *titration* term, the one piece of ozone chemistry that does not require observing VOCs, and it must not be generalized to regional photochemical production.

5.2 Gate 2: unobserved planetary-boundary-layer confounding

The diurnal cycle of the planetary boundary layer (PBL) is a powerful *unobserved common cause*. At night and in winter the PBL collapses, trapping fresh emissions in a shallow volume and simultaneously raising the measured concentrations of all primary, co-emitted species; by day it deepens and dilutes them together. This single latent variable induces strong positive correlations among co-emitted pollutants that have nothing to do with any direct inter-pollutant causal mechanism, violating the causal sufficiency assumption (A4) of (3). The empirical signature is exactly what we observe: the particulate and combustion edges of Figure 4 are stable in the pollutant-only graph but collapse once meteorological proxies of mixing are included. A spuriously "discovered" CO $\to$ $PM_{2.5}$ edge is better read as two species responding to a common,

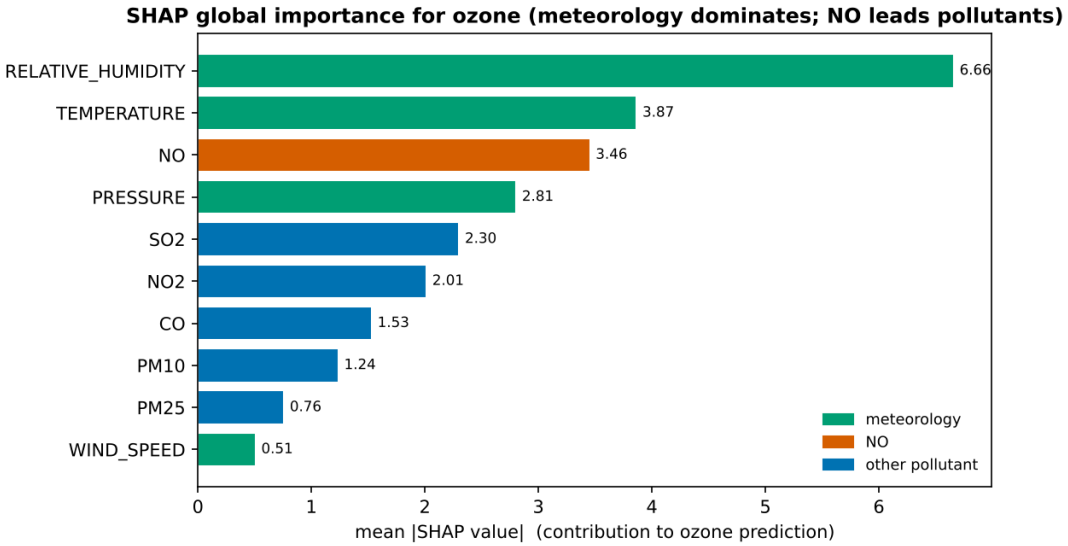


Figure 6: Global SHAP importance (mean |SHAP|) for ozone estimation. Meteorology dominates; NO is the leading chemical covariate, consistent with Figure 4. Per-sample attributions were not retained, so a beeswarm is not shown.

unmeasured mixing volume. Mitigation requires either direct PBL-height observations or explicit meteorological deconfounding [4, 6, 7].

5.3 Gate 3: the concentration–emission fallacy

This is the decisive gate for policy. The optimizer of Lemma 1 acts on *standardized ambient concentrations*: a recommendation such as "$z_{\mathrm{NO_2}} = z_{\min} = -2\sigma$" is a point on a correlational surface, not an emission control. Two failures follow. First, there is no validated map from this concentration vector back to a feasible set of sectoral emission reductions; the inverse problem is precisely the ill-posed source attribution that (1) would have to solve. Second, the search can select physically impossible states: for a right-skewed, non-negative pollutant whose standardized mean sits near zero, a target of -2σ can correspond to a *negative physical mass concentration*. Formally, if $C = \mu + \sigma z$ with $\mu \lesssim 2\sigma$ (as holds for several species in Table 1), then $z = -2$ implies $C < 0$. An "optimal policy" that prescribes negative pollutant mass is a reductio: it exposes that the objective never encoded emissions, feasibility, or benefit. Optimizing a concentration response surface and reporting the result as a control strategy is a category error, independent of which optimizer is used.

5.4 Gate 4: leakage-aware evaluation

Finally, no skill claim survives without an autocorrelation-aware protocol. As Table 2 shows, the same model and data yield R^2 from 0.41 to 0.22 depending solely on the split. Random k-fold over rows of spatio-temporal telemetry is not a valid estimate of out-of-sample skill [12, 25]; blocked-temporal and leave-groups-out protocols, benchmarked against persistence and climatology, are the minimum standard.

5.5 Recommendations

For researchers and regulators deploying observational environmental AI we recommend: (i) report blocked and grouped validation with persistence and climatology baselines, never a single random-split number; (ii) treat discovered graphs as hypotheses and test every edge against meteorological confounding and bootstrap stability before any causal language; (iii)

verify measurement units and provenance against authoritative sensor metadata, not value heuristics; (iv) reserve "source apportionment" for receptor models (2) with speciation and uncertainties; and (v) never present an optimizer over ambient concentrations as an emission-control recommendation without a validated, feasibility-constrained emissions-to-concentrations model.

6 Conclusion

We audited a complete prediction-to-policy air-quality pipeline on 1,180,551 station-hours from 204 Indian urban monitors and found that its most consequential claims do not survive scrutiny. Empirically, spatio-temporal autocorrelation inflates apparent predictive skill from an honest $R^2 \approx 0.22$–0.30 to a leaky 0.41, and the learned model is beaten by trivial persistence for temporal gap-filling; pollutants add only $\Delta R^2 \approx 0.12$ beyond meteorology. Structurally, linear causal discovery recovers a single robust edge—localized NO–O_3 titration—rather than any regional photochemical mechanism, a ceiling imposed by the structural absence of VOCs and sharpened by a verified ppb/μg m^{-3} unit-provenance defect in three species. Algorithmically, we proved (Lemma 1) that the true Pareto frontier of the fitted linear surrogate is a closed-form line segment, against which 100% of the heuristic NSGA-II policies are dominated (GD = 0.084, HVE = 6.6%).

Taken together these results motivate the four-part identification gate of Section 5 and a deliberate reframing. Rather than an over-claimed causal source-apportionment and control-optimization system, the artifact is most honestly and usefully released as an *open, leakage-safe benchmark* for multi-pollutant air-quality machine learning: a real national-scale dataset, a reproducible evaluation protocol, a baseline ladder that future methods must beat, one chemically interpretable hypothesis, and an explicit account of what each stronger claim would require. All code, data, seeds and analysis scripts are released to this end.

Future work. The productive path forward couples, rather than replaces, mechanism and data. We see three directions: (i) *hybrid modelling*, in which distributed observational surrogates are coupled to, and constrained by, regional three-dimensional chemical transport models (e.g. WRF-Chem, CMAQ) so that interventions are simulated through (1) rather than fit on a correlational surface; (ii) *speciation*, through low-cost VOC and carbonaceous-aerosol sensor arrays or satellite HCHO/NO_2 proxies, to pass Gate 1 and enable genuine ozone-regime and receptor-model analysis; and (iii) *identified causal inference*, using documented interventions or natural experiments with meteorological deconfounding and pre-registered estimands. Until an analysis passes the identification gate, the responsible contribution of observational environmental AI is a well-evaluated predictive baseline and an honest map of its own limits.

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Author contributions

S. Krishiykeishav, Pratyush Amit Sharma, Shreerag Venugopalan Nair, Jayshil Vinodbhai Prajapati and Ryan Vinod jointly designed the study, implemented the pipeline and audit, and wrote the manuscript.

Reproducibility statement

All code, the committed telemetry dataset, fixed random seeds, per-phase manifests, and the scripts that regenerate every number, table and figure in this manuscript are available in the project repository on the `review/scientific-audit` branch. Every result is reproducible offline without a cluster; the audit scripts fix `SEED = 42`.

Data availability

Ambient measurements were obtained from the OpenAQ v3 API [18]; meteorological reanalysis from Open-Meteo [36]. Derived analysis artifacts are committed to the repository.

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